

RATE GYRO ASSEMBLY (RGA) — REMOVE & REPLACE

1.480
RGA R&R
(ISS EVA SYS / FIN)

EV1 lead / EV2 assist | Worksite: S0 truss, RGA-2 ORU bay | Est. task time 0:55 | Crew lock egress hatch

OBJECTIVE

Remove failed RGA-2 orbital replacement unit (ORU) and install spare RGA from ETB. RGA provides rate data to the GN&C MDM for attitude control. Failed unit is power-isolated by ground prior to egress. Capture torque values and connector mate status; downlink via voice and helmet-cam. Steps marked **[GA]** are get-ahead tasks — perform only if ahead of timeline and consumables permit.

WARNING: Verify RGA-2 RPCM channel OPEN (ground-confirmed) before breaking any connector. Do not actuate without GO from MCC-H. Sharp-edge hazard on ORU guide rail — maintain glove awareness.

1.0 WORKSITE SETUP & SAFETY

EV / GMT	ACTION	EXPECTED INDICATION
EV1 —	1.1 Translate to S0 RGA-2 bay. Install APFR in receptacle S0-04 , pitch 3 , roll F , yaw 6 .	<i>APFR locked; black-on-black alignment.</i>
EV1 —	1.2 Ingress APFR. Tether shorten to worksite. Configure BRT for RGA mass.	<i>Two tethers verified — one structural, one local.</i>
EV2 —	1.3 Stage ETB at S0 handrail HR-0142 . Retrieve spare RGA, verify P/N on ORU label.	<i>P/N matches: RGA-1F2 / S/N 1007.</i>
IV —	1.4 CALL MCC-H: confirm RGA-2 RPCM open and unit power-isolated. Standby for GO.	<i>MCC-H: "You have a GO for RGA R&R."</i>

↘ crew note (in flight): "RPCM confirmed open 14:22 GMT — KH"

2.0 REMOVE FAILED RGA-2

EV / GMT	ACTION	EXPECTED INDICATION
EV1 —	2.1 Demate connector J1 (power). Rotate collar CCW $\approx 1/4$ turn to soft-dock, then full demate.	<i>Connector free; no pin damage. Stow on bib tether.</i>
EV1 —	2.2 Demate connector J2 (data). Verify backshell strain-relief clear.	<i>J2 free; dust cap installed on bay receptacle.</i>
EV1 —	2.3 Engage PGT: Setting 1 (CCW), Torque 30 ft-lb, RPM 30 . Break 4 captive bolts (B1–B4), 7 turns ea.	<i>PGT reads 4 of 4 bolts backed out; white index paint visible on each bolt.</i>
EV1 —	2.4 Grasp RGA-2 handhold. Extract along guide rail +Z until clear of bay.	<i>ORU clears bay without bind; rail witness marks aligned.</i>
EV2 —	2.5 Receive RGA-2. Temp-stow on ETB. Record bolt torque values for downlink.	<i>Failed ORU restrained; logged. [GA] photo-doc connector pins.</i>

CONTINGENCY — IF BOLT WILL NOT RELEASE (2.3):

- STOP. Do not exceed 38 ft-lb (bolt yield). Reseat PGT square on bolt head, retry at **RPM 10**.
- If still seized after 2 attempts → switch to manual ratchet, 1/8-turn increments.
- If seized bolt > 1 → CALL MCC-H, hold task, proceed to **get-ahead 5.0** while ground assesses.
- Track suit consumables; abort task at **BINGO O2** call.

3.0 INSTALL SPARE RGA

EV / GMT	ACTION	EXPECTED INDICATION
EV2 —	3.1 Hand spare RGA to EV1. Remove receptacle dust cap; stow cap on ETB.	<i>Cap stowed; receptacle clean, no FOD.</i>
EV1 —	3.2 Align spare on guide rail. Insert –Z until soft-dock; verify witness marks.	<i>ORU seated flush; index lines aligned within 1/8 in.</i>

EV1 —	3.3 PGT: Setting 2 (CW), Torque 30 ft-lb. Drive B1–B4 in star pattern to full seat.	<i>PGT 4 of 4 at target torque; audible/visual torque-out on each.</i>
EV1 —	3.4 Mate J2 (data) first, then J1 (power). Collar CW to hard-stop; verify backshell flush.	<i>Both connectors mated; no gap at collar witness band.</i>
IV —	3.5 CALL MCC-H: ready for RPCM close / aliveness test. Standby on connectors.	<i>MCC-H: "RGA aliveness nominal, rate data good."</i>

↘ crew note (in flight): "aliveness GO 15:05 GMT — rates tracking, no flags"

4.0 CLOSEOUT

EV / GMT	ACTION	EXPECTED INDICATION
EV1 —	4.1 Visually inspect ORU seat, connectors, bolts. Confirm no protruding hardware.	<i>Worksite clean; bolts seated; no FOD released.</i>
EV1/EV2 —	4.2 Account for all tools (PGT, ratchet, caps, tethers). Tool inventory to IV.	<i>Tool count complete — matches egress manifest.</i>
EV2 —	4.3 Restrain failed RGA-2 in ETB for return. Translate ETB toward A/L.	<i>Failed ORU secured for ingress; log entry complete.</i>

5.0 GET-AHEAD TASKS [GA] — only if ahead of timeline

- **5.1** Photo-document adjacent RGA-1 connector torque stripes for trend tracking.
- **5.2** Image debris strikes on S0 handrails HR-0140 thru HR-0145; note any frayed MLI.
- **5.3** Verify tightness of WIF caps at S0-05/06 (prior-EVA open item).
- **5.4** Stage spare bolt/connector kit at HR-0142 for next planned EVA.

VOICE / DISPLAY LOG MAPPING (demo aid). Each **EXPECTED INDICATION** cell is a confirmable event: the crewmember vocalizes the actual reading, IV/MCC-H verifies against this card, and the call is timestamped. For a hands-free HUD logging system, each step ID (e.g. 2.3) is the log key; the expected-cue text is the assertion to confirm or flag off-nominal. Contingency box = the decision node the display surfaces when an assertion fails.